Report for the final assignment for Data Analytics and Statistics Seminar

Question 1.

- What does one row represent? One row represents one observation of 442 patients accessed from the source: www4.stat.ncsu.edu/~boos/var.select/diabetes.tab.txt

- Which variable is the outcome? The variable that is the outcome is the Y

value that was named after as 'progression'

- Which variables are predictors? All the measurements from the patients could be potential predictors. Predictors are defined as the independent variables, that we tried to fit the linear regression model, to see if can predict the output dependent variable. So, in our case potential predictors are: age, sex, bmi (body mass index), bp (blood pressure), s1, s2, s3, s4, s5, s6 (6 serum measurements)

- Which variables are numerical, and which are categorical in meaning? All the variables are numerical except in the sex case, which is a categorical one. Noted: in our case there aren't used the conventional 0 and 1, but the 1 and 2 values for some reason I am not aware of.

- Are there missing values, duplicated rows or obviously problematic values? Based on the output provided by the source R code, there are not any missing values or duplicated rows, based on some histogram graphs, maybe we can assess that there are some outliers, that influence our analysis and it is a limitation of our study, as we didn't exclude any of the data.

- Why do health-related data require careful and responsible interpretation? Behind health-related data there are ethical concerns and consideration, based on the GDPR legislation in Europe, after the right of individuals to secure their personal data. In health-related data set during the collection of the observation one must be careful, so no people identified directly or indirectly. Also, there higher stakes general in your assumptions during the statistical analysis, with big effect in the final decisions. Note: that in this case there is for educational purposes, and the number of observations are limited.

Question 2.

- What do the centre and spread of the main variables look like? Based on the heatmap of the correlation matrix, for a preliminary feature selection, the main variables are assigned to: bmi, bp and s5 as they have higher correlation with the progression variable as we can see in the image 1. The center of the variables can be seen in the image 2. The means of BMI, BP and S5 are 26.4, 94.6 and 4.64 correspondingly.

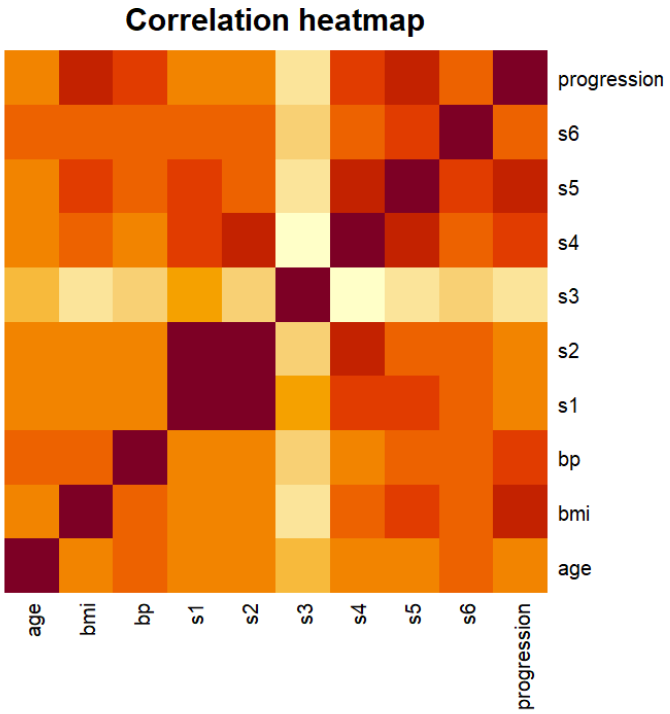


Image 1. Correlation Heatmap of the different variables of dataset

```
      bmi              bp               s5        
 Min.   :18.0   Min.   : 62.0   Min.   :3.26  
 1st Qu.:23.2   1st Qu.: 84.0   1st Qu.:4.28  
 Median :25.7   Median : 93.0   Median :4.62  
 Mean   :26.4   Mean   : 94.6   Mean   :4.64  
 3rd Qu.:29.3   3rd Qu.:105.0   3rd Qu.:5.00  
 Max.   :42.2   Max.   :133.0   Max.   :6.11  
```

Image 2. Important descriptive metrics of the main variables of our model

- Are the distributions approximately symmetric or clearly skewed?

After plotting the histograms for the three main variables, we found that the distributions are approximately normal. There is a little bit skewness in the BMI distribution, but it isn't important as mean is close to the value of median (mean is sensitive to extreme outliers, skewed data). You can see the histograms in the images 3, 4 and 5.

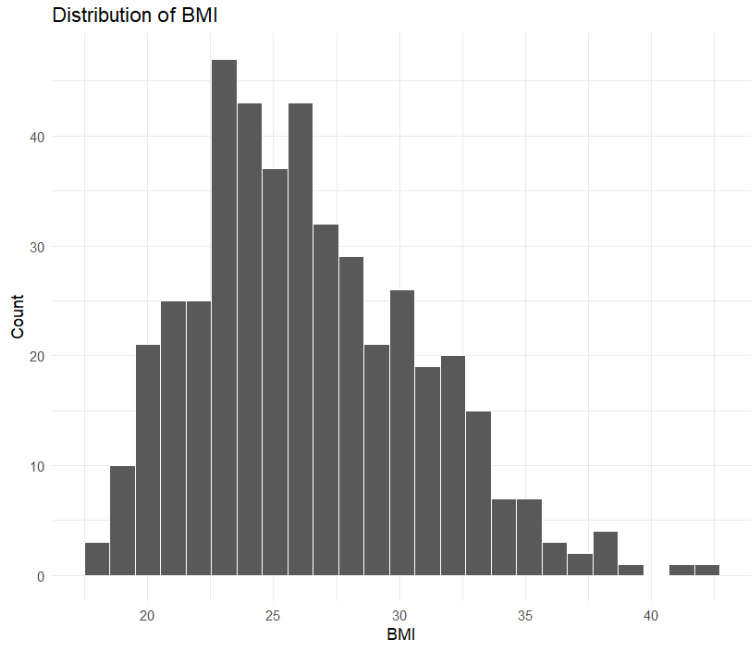


Image 3. Histogram of BMI

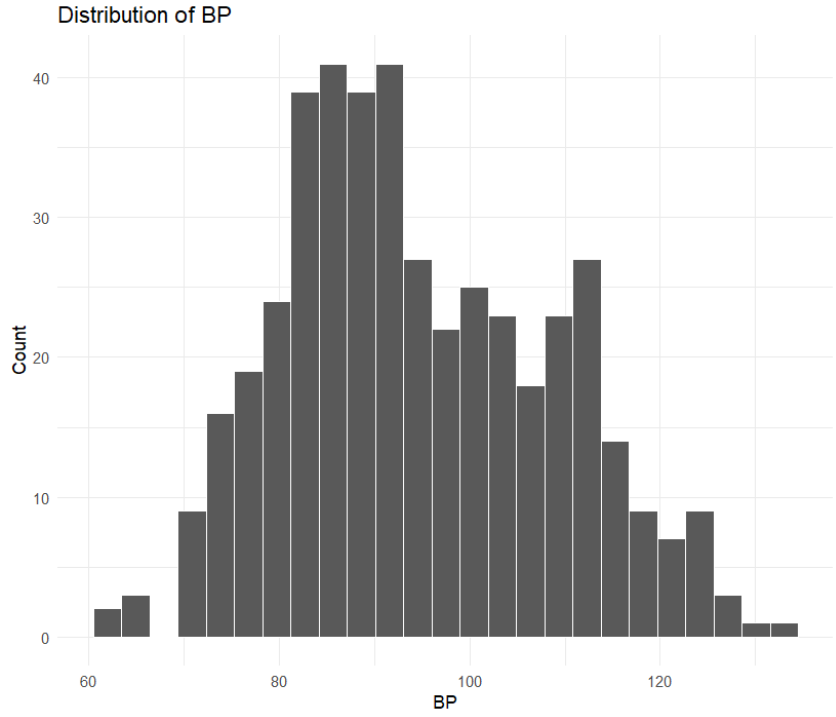


Image 4. Histogram of BP

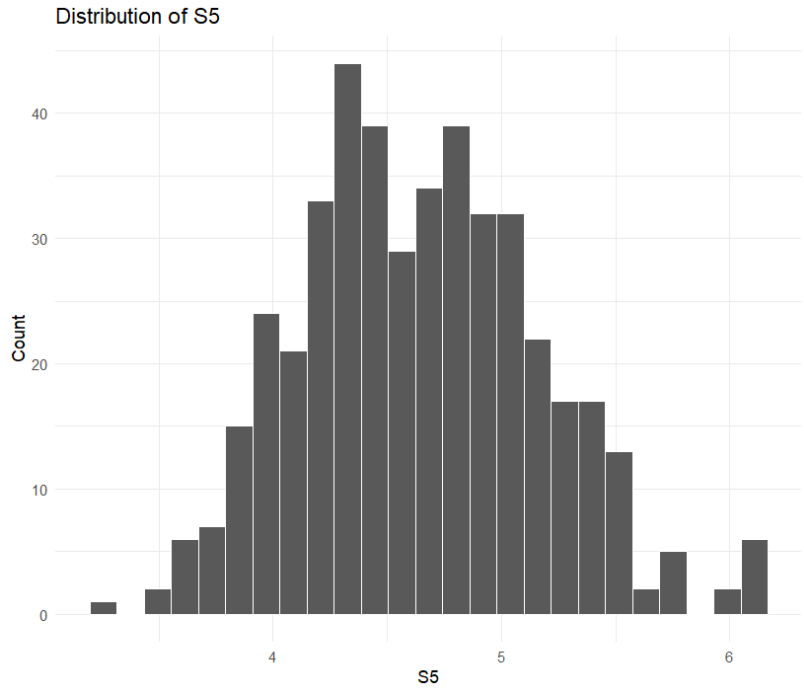


Image 5. Histogram of S5

- Which features of the data matter before modelling begins? Important features that matter for the OLS fitting, are mainly the mean, which as we shown is close to the median value, meaning the are not that extreme outliers, even if we see some skewed observation, so there isn't a need for preprocessing of the data to filter outliers. Even though with more time we could study the behaviour of the system when some values are excluded.

Question 3.

- Which predictors are most strongly associated with progression? We already presented the correlation heatmap, which shows that maybe there is a strong correlation effect between the variables bmi, bp and s5 with progression

- Which predictors are strongly related to each other? Looking into the image 1, there is a characteristic pair, that seems to be strongly correlated to each other and maybe show overlapping information, that I would probably avoiding putting both in a fitting model, as I except some collinearity effect, even though we more time should definitely be proved. The pair I am commenting on is the s1-s2 variables.

- Which variables look most promising for regression analysis, and why?

In image 6. there are provided the progression correlation values

bmi	s5	bp	s4	s6	s1	age
0.5865	0.5659	0.4415	0.4305	0.3825	0.2120	0.1879

Image 6. Progression Correlation Values

Question 4.

- Write the fitted equation.

The fitted equation based on the bmi regression linear model is provided:

Progression = 10.233*bmi -117.773

- Interpret the slope in context.

For each increase in the unit of bmi, the progression diabetes' score is increased by 10.233 units.

- Does the intercept have a useful real-world interpretation? No, in this situation that we know the range of bmi is in values 18 – 42.2, it doesn't make sense to have bmi = 0, the minimum value the model makes sense is the 18 bmi value.

- What does R-squared mean here? The R-squared in this model explain the variability in the regression scored based on the bmi for 34.2%, that isn't generally very strong, so definitely we have to add more variables to fit a multilinear model.

Question 5.

- Interpret at least two coefficients from the multiple model.

```
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) -334.881     25.878  -12.94  < 2e-16 ***
bmi            6.500      0.697    9.32  < 2e-16 ***
bp             0.903      0.217    4.17  3.7e-05 ***
s5            49.577      5.890    8.42  5.6e-16 ***
```

Image 7. Slopes and other important metrics for the fitted multimodal based on bmi, bp and s5 variables

It seems there is more strongly effect in the output from s5 and bmi as strongly was indicated by the progression correlation values, with an increase of 1 unit of bp, there isn't that much addition to the diabetes progression, although isn't this the case for s5 serum measurement. It seems, that this biomarker is very important.

- Compare the simple and multiple models.

- Do not assume that the model with the larger R-squared is automatically better in every respect. Indeed, adding more predictors should increase automatically the R-squared value, but after the adjustment of the R-squared still remained constant in respect of the R-squared value, so the variables adding provide added value to the model as the R-squared valued went from 0.344 for the first model to 0.48 to the second one. But it seems that bp doesn't have that strong effect, so in our next step we tried to remove from the model to compare the models. The R-squarede moved towards the value of ~0.46 which is very close to 0.48, so for having a simpler model

we would prefer this third option. Theoretically, we have the view that age plays an important role into the development of the diabetes. Let's see. It seems that age has zero effect to the progression of the diabetes!

```
term        estimate std.error statistic  p.value conf.low conf.high
<chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
(Intercept)  -300.      24.9      -12.0  5.25e-29  -349.     -251.
bmi             7.28     0.684     10.6  1.17e-23     5.93      8.62
s5             56.1      5.79       9.69 3.04e-20    44.7      67.4

glance(multiple_model)
A tibble: 1 x 12
r.squared adj.r.squared sigma statistic  p.value    df logLik   AIC   BIC deviance df.residual
    <dbl>         <dbl> <dbl>     <dbl>    <dbl> <dbl>  <dbl> <dbl> <dbl>    <dbl>       <int>
    0.459         0.457  56.8      187. 2.25e-59     2 -2411. 4830. 4847. 1416694.         439
```

Image 8. Third Model based only on bmi and s5 variables

```
(Intercept) -301.        25.2     -11.9   1.12e-28 -350.      -251.
bmi            7.26       0.687    10.6   1.95e-23    5.91       8.61
age            0.0502     0.215     0.233 8.16e- 1   -0.373      0.473
s5            55.8        5.93      9.40  2.97e-19   44.1       67.4
```

Image 9. Fourth Model based also on age

Question 6.

- Comment on the residuals-versus-fitted plot.

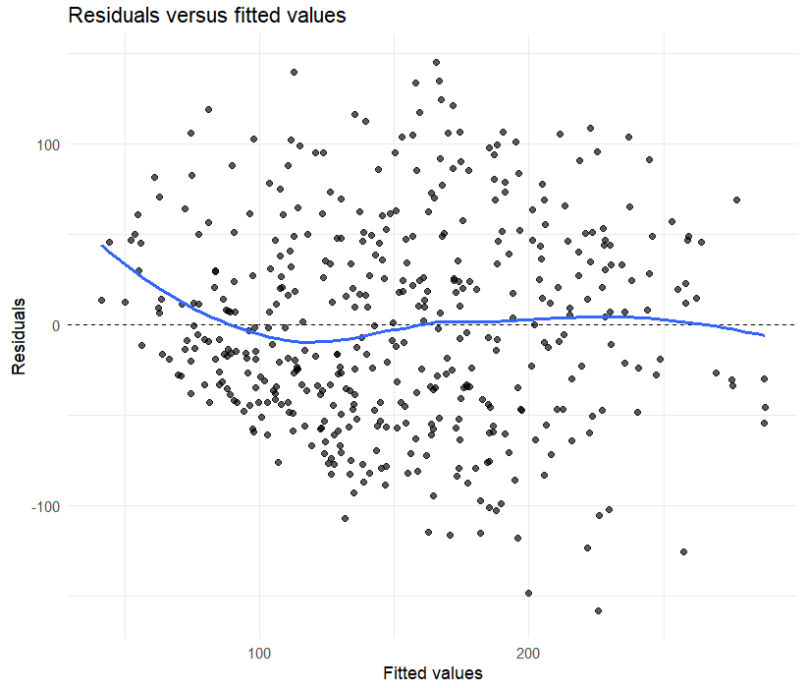


Image 10. Residuals-versus-fitted plot

Generally, the residual error, show a good distribution over fitted values, as the sum of those is approximately constant close to 0, although the scatteplot shows a cluster, that could be a warning to check also for non-linear effect. Note: This is a limitation of this study, we more time we would add more variables and check with polynomial or other non-linear models, if there are added effects to predict the progression.

- Use VIF and the correlation matrix to discuss multicollinearity.

As we already discussed the multicollinearity based on the correlation matrix, here we provide evidence of the values of VIF we used in the fourth model. The values are very low, so there isn't a collinearity based on those pair variables.

```
  bmi   age    s5
1.256 1.085 1.308
```

Image 11. VIF Values of Fourth Fitted Model.

- For at least two important coefficients, discuss the estimate and either

the confidence interval, t-value or p-value.

Generally the values of the important variables as you can see in the previous images are not that big, although p-value in most cases it is very negligible. So, there is strong evidence against null-hypothesis.

- Distinguish statistical significance from practical importance.

- State the main statistical weaknesses or uncertainties in the analysis. We need to check all the variables and maximize the adjusted R-squared, check the multicollinearity effects of the pair s1-s2, check non-linear effects, small sample size, there isn't ofcourse causality, but BMI and S5 seems to play an important part to the progression of diabetes

Question 7.

- Is this dataset experimental or observational? This dataset is observational.

- Can the regression coefficients be interpreted causally? No, they can't even though maybe there is a strong theoretical foundation behind bmi and metabolic syndromes, that could establish the causal relationship between at least of the diabetes and bmi

- Could there be confounding variables that are not included in the dataset? Yes, maybe the already there are see s1-s2 pair

- Could measurement issues affect the results? Ofcourse, skewed data, or just measurement issues can change the mean which as an important feature for the OLS fitting of the model

- Would the results necessarily generalise to another population? No, dataset is too smal, 442 patients.

- Which conclusions are supported by the evidence? BMI and S5 have the most strong effect associating them as potential risks for the diabetes progression. Also, even though the age was expected to have an impact, this was completely discarded as an assumption, even though it could be cause of the small dataset, of maybe it is an confounder, although age could be linked with the other measurements too.